Binary Logistic Regression Case Study: Marketing Campaign Response Model

Background: The large skin-clinic in the city of “XXX” offers variety of products and services to customers. The marketing campaign was launched for introducing a new product.

The data shows response to the marketing campaign alongwith demographic and transaction data of customers.

The objective is to identify factors driving response to the marketing campaign.

Import datasets and create masterdata

```
purchase_data1<-read.csv("Purchase Data 1.csv",header=T)
purchase_data2<-read.csv("Purchase Data 2.csv",header=T)
response_data<-read.csv("Response Data.csv",header=T)
library(dplyr)

masterdata<-full_join(response_data,purchase_data1,by="Custid")
masterdata<-full_join(masterdata,purchase_data2,by="Custid")
```

Convert numerically coded categorical variables into “factors”

```
masterdata$Age<-as.factor(masterdata$Age)
masterdata$Gender<-as.factor(masterdata$Gender)
masterdata$MS<-as.factor(masterdata$MS)
masterdata$Pre_Month<-as.factor(masterdata$Pre_Month)
```

Display first 6 rows

```
head(masterdata)
```

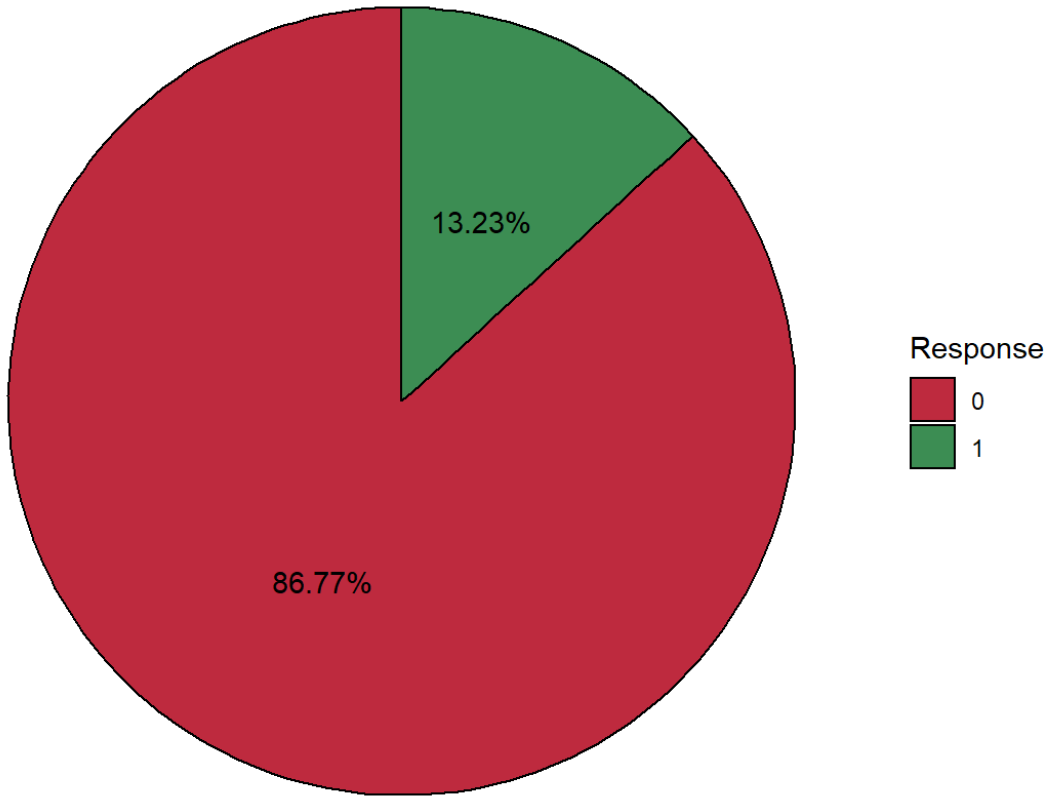
```
  Custid Age Gender MS Response Pre_Month N_Products N_Service BillAmt_1
1      1   2      1  2        1         2         15        24     12.34
2      2   2      2  1        0         1         22        29     18.65
3      3   1      1  2        0         1         17        21      7.22
4      4   2      1  1        0         2         18        22      6.15
5      5   2      2  1        0         1         31        35     20.64
6      6   1      2  1        0         2         28        31     12.44
  BillAmt_2 BillAmt_3
1     13.26      5.88
2      2.12      5.13
3      3.31      3.65
4      2.95      2.34
5      2.67      4.07
6      3.06      2.57
```

Data Description

Column name	**Column desciption**
Custid	Customer Identification Code
Age	Age group(1:<32,2:32-48,3:>48)
Gender	Gender(1:Female,2:Male)
MS	Marital Status(1: Not Married,2:Married)
BillAmt_1.BillAmt_2.BillAmt_3	Bill Amounts (USD) for Last Three Purchases
Pre_Month	Purchase in Previous Week(1: Purchased in Previous Week,2: No purchase in Previous Week)
N_Products	Number of unique products purchased in One Year
N_Service	Number of unique services purchased in one year

Visualize Campaign Response Rate

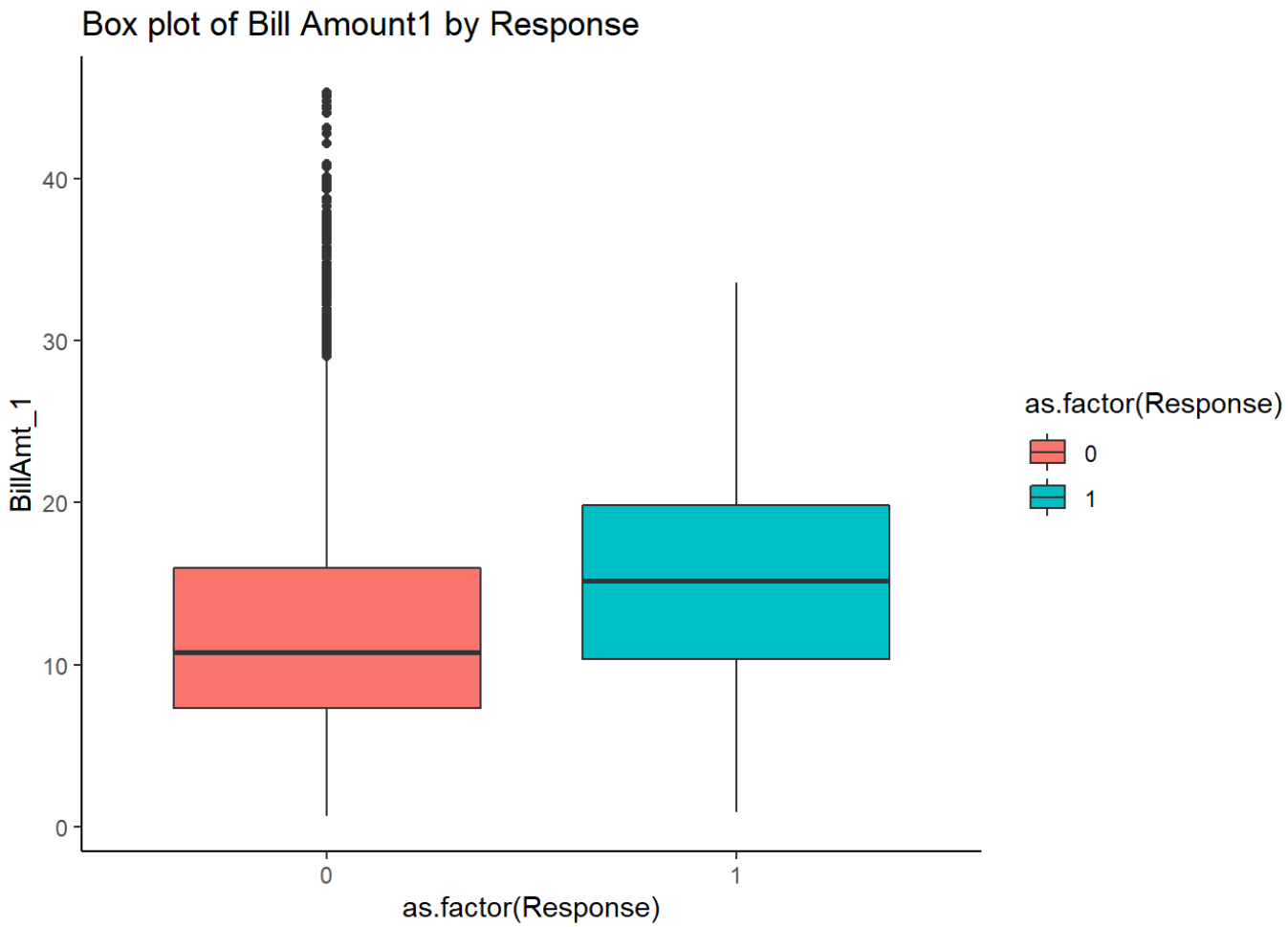
```
library(ggplot2)

response<-data.frame(table(masterdata$Response))
names(response)<-c("Response","Count")
response$Perc<-paste(round((response$Count/sum(response$Count))*100,2),"%",sep = "")

ggplot(response, aes(x = "", y = Count, fill = Response)) +
  geom_col(color = "black") +
  geom_text(aes(label = Perc),
            position = position_stack(vjust = 0.5)) +
  coord_polar(theta = "y") +
  scale_fill_manual(values = c("#BE2A3E", "#3C8D53"))+
  theme_void()
```



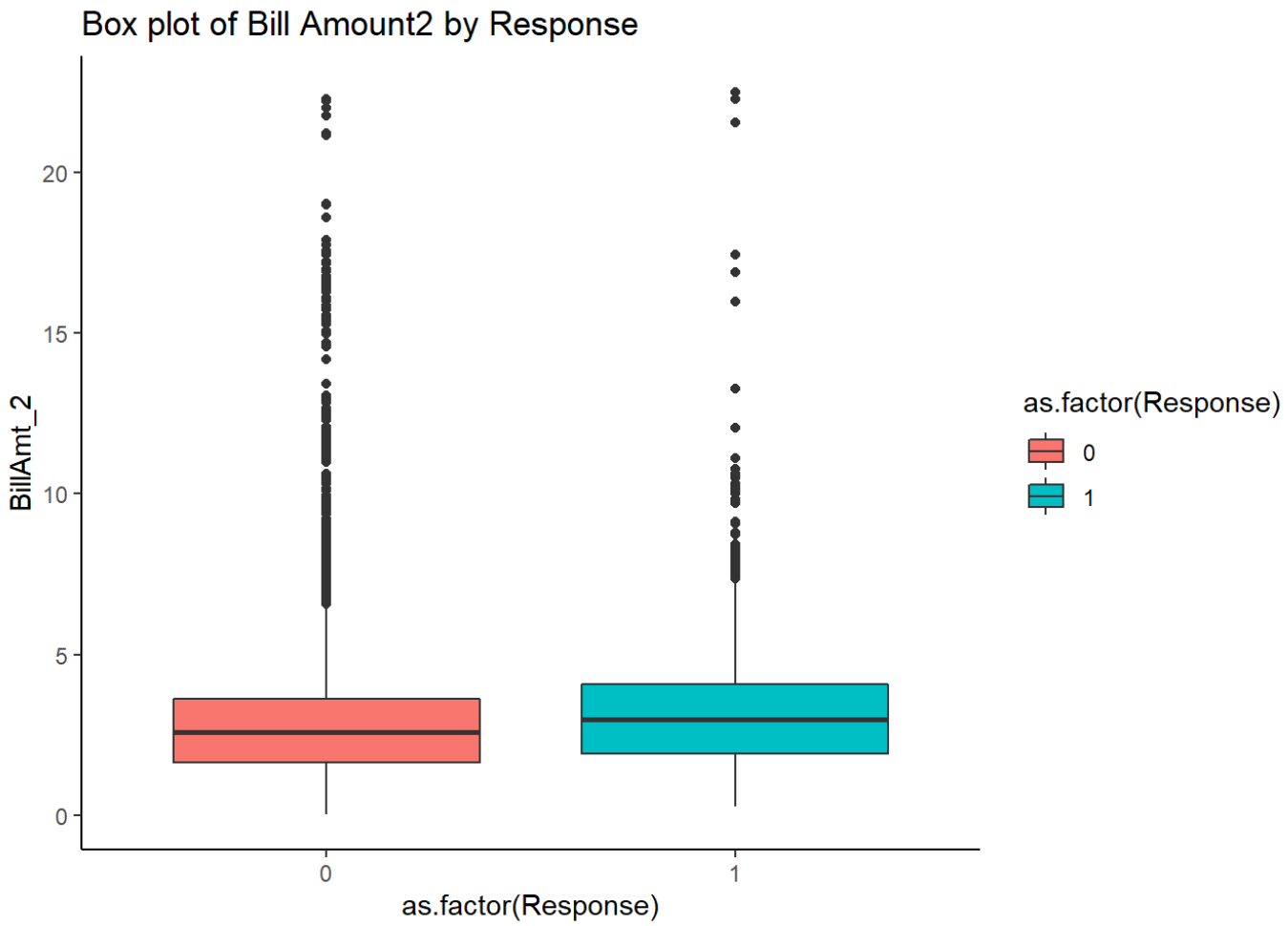
Comment: Approximately 13% of customers responded positively to the campaign

Box plot of Bill Amounts by Response

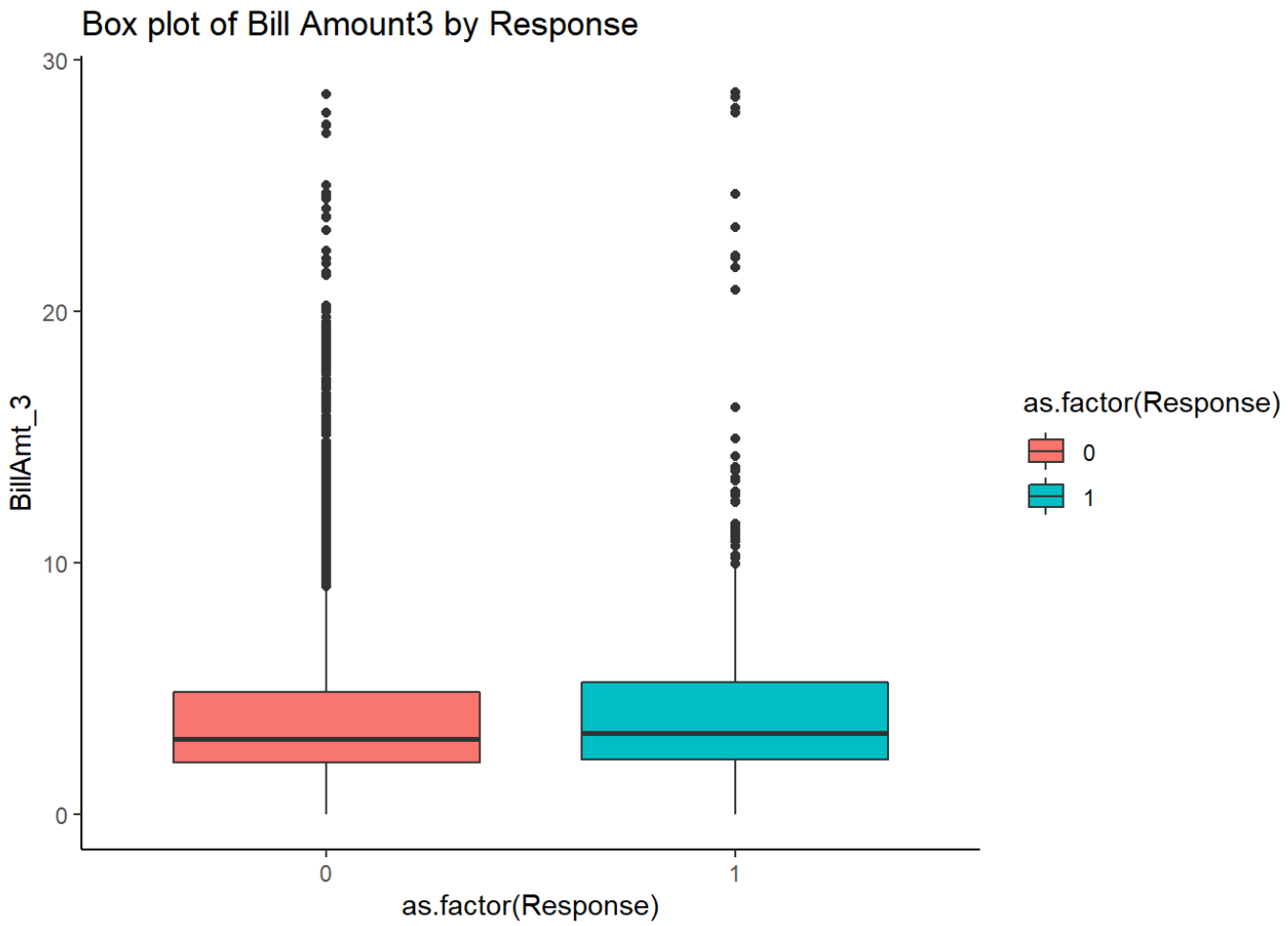
```
ggplot(masterdata, aes(x=BillAmt_1, y=as.factor(Response),fill=as.factor(Response))) +
  ggtitle("Box plot of Bill Amount1 by Response")+
  geom_boxplot()+ coord_flip()+theme_classic()
```




```
ggplot(masterdata, aes(x=BillAmt_2, y=as.factor(Response),fill=as.factor(Response))) +
  ggtitle("Box plot of Bill Amount2 by Response")+
  geom_boxplot()+ coord_flip()+theme_classic()
```




```
ggplot(masterdata, aes(x=BillAmt_3, y=as.factor(Response),fill=as.factor(Response))) +
  ggtitle("Box plot of Bill Amount3 by Response")+
  geom_boxplot()+ coord_flip()+theme_classic()
```



Comment: Higher bill amounts in last 2 transactions are observed for customers with positive response

Develop Campaign Response Model using Binary Logistic Regression

Split original data into training and testing data sets

```
library(caret)
masterdata$Response<-as.factor(masterdata$Response)
index<-createDataPartition(masterdata$Response,p=0.7,list=FALSE)
traindata<-masterdata[index,]
testdata<-masterdata[-index,]
```

Binary Logistic Regression using glm function

```
blr_model<-glm(Response~Age+Gender+MS+Pre_Month+N_Products+N_Service+BillAmt_1+BillAmt_
2+BillAmt_3,
                    data=traindata,family = binomial)
```

Display parameter estimates with other model statistics

```
summary(blr_model)
```

```


Call:
glm(formula = Response ~ Age + Gender + MS + Pre_Month + N_Products +
    N_Service + BillAmt_1 + BillAmt_2 + BillAmt_3, family = binomial,
    data = traindata)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.5854  -0.5876  -0.4001  -0.2195   2.6778

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -7.786534   0.398695 -19.530  < 2e-16 ***
Age2        -0.243687   0.108734  -2.241    0.025 *
Age3        -0.094811   0.107105  -0.885    0.376
Gender2      0.006044   0.088912   0.068    0.946
MS2         -0.084680   0.088957  -0.952    0.341
Pre_Month2   0.016065   0.088971   0.181    0.857
N_Products   0.152382   0.010600  14.375  < 2e-16 ***
N_Service    0.034020   0.007858   4.329 1.50e-05 ***
BillAmt_1    0.037330   0.007543   4.949 7.46e-07 ***
BillAmt_2    0.181133   0.025150   7.202 5.93e-13 ***
BillAmt_3    0.016312   0.018873   0.864    0.387
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3826.5  on 4893  degrees of freedom
Residual deviance: 3366.9  on 4883  degrees of freedom
AIC: 3388.9

Number of Fisher Scoring iterations: 6
```

Comment: “Age”,“Gender”,“MS”,“Pre_Month” and “BillAmt_3” are the insignificant variables.

Rerun the model after removing the insignificant variables

```
blr_model<-glm(Response~N_Products+N_Service+BillAmt_1+BillAmt_2,
                   data=traindata,family = binomial)
summary(blr_model)
```

```
Call:
glm(formula = Response ~ N_Products + N_Service + BillAmt_1 +
    BillAmt_2, family = binomial, data = traindata)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.6156  -0.5878  -0.4022  -0.2206   2.6556

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -7.837382   0.374591 -20.922  < 2e-16 ***
N_Products   0.149708   0.010224  14.643  < 2e-16 ***
N_Service    0.033406   0.007772   4.298 1.72e-05 ***
BillAmt_1    0.040209   0.006767   5.942 2.82e-09 ***
BillAmt_2    0.188932   0.023633   7.994 1.30e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3826.5  on 4893  degrees of freedom
Residual deviance: 3373.7  on 4889  degrees of freedom
AIC: 3383.7

Number of Fisher Scoring iterations: 6
```

Comment: The 4 significant purchase related variables have positive regression coefficients. Higher values are associated with positive campaign response.

Check for multicollinearity using vif function

```
library(car)
vif(blr_model)
```

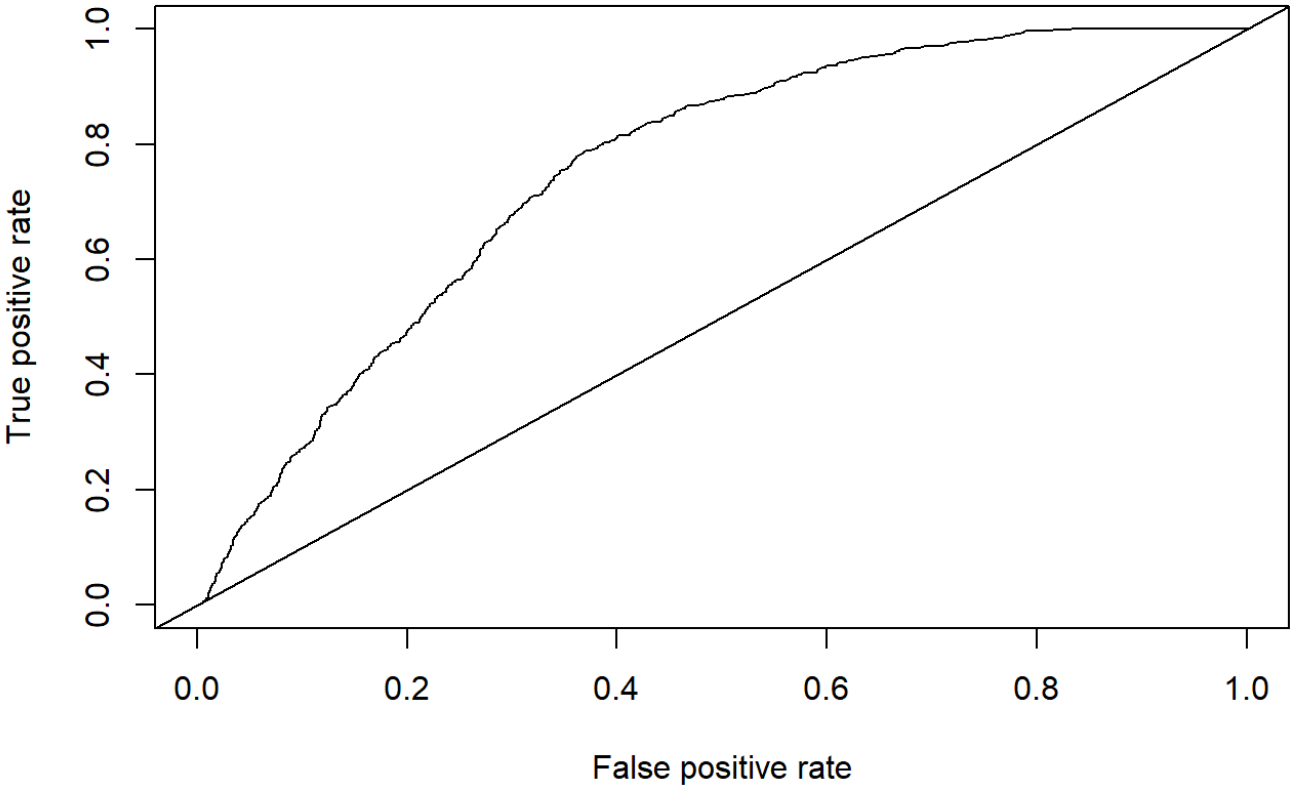
```
N_Products  N_Service  BillAmt_1  BillAmt_2
  1.373462   1.091970   1.310203   1.753729
```

Comment: It is observed that no variable has high vif

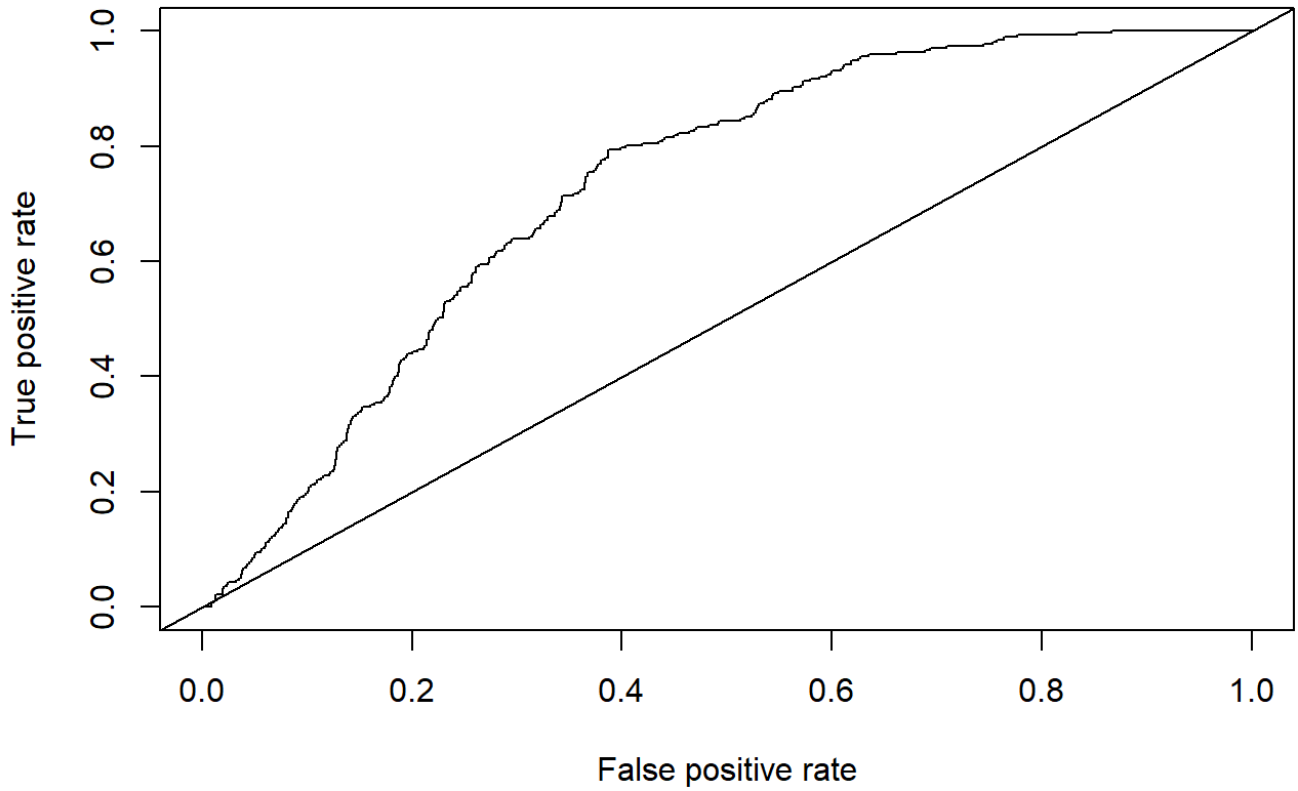
Hence the problem of multicollinearity does not exist

Evaluate model performance by ROC Curve using ROCR package

```
library(ROCR)
traindata$predprob<-fitted(blr_model)
predtrain<-prediction(traindata$predprob,traindata$Response)
perftrain<-performance(predtrain,"tpr","fpr")
plot(perftrain)
abline(0,1)
```




```
testdata$predprob<-predict(blr_model,testdata,type='response')
predtest<-prediction(testdata$predprob,testdata$Response)
perftest<-performance(predtest,"tpr","fpr")
plot(perftest)
abline(0,1)
```



Checking area under the ROC curve

```
auctrain<-performance(predtrain,"auc")
auctrain@y.values
```

```
[[1]]
[1] 0.7541103
```

```
auctest<-performance(predtest,"auc")
auctest@y.values
```

```
[[1]]
[1] 0.7315719
```

Model Validation: Holdout Method using confusionMatrix

Finding optimum threshold for classification by maximizing sensitivity + specificity

```
sstrain <- performance(predtrain, "sens", "spec")
best_threshold <- sstrain@alpha.values[[1]][which.max(sstrain@x.values[[1]]+sstrain@y.v
alues[[1]])]
best_threshold
```

```
[1] 0.1292341
```

Confusion matrix for train and test data

```
traindata$predY<-as.factor(ifelse(traindata$predprob>best_threshold,1,0))
confusionMatrix(traindata$predY,traindata$Response,positive="1")
```

```
Confusion Matrix and Statistics

          Reference
Prediction    0    1
         0 2680  138
         1 1566  510

               Accuracy : 0.6518
                 95% CI : (0.6383, 0.6652)
    No Information Rate : 0.8676
    P-Value [Acc > NIR] : 1

                  Kappa : 0.2163

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.7870
            Specificity : 0.6312
         Pos Pred Value : 0.2457
         Neg Pred Value : 0.9510
             Prevalence : 0.1324
         Detection Rate : 0.1042
   Detection Prevalence : 0.4242
      Balanced Accuracy : 0.7091

       'Positive' Class : 1
```

```
testdata$predY<-as.factor(ifelse(testdata$predprob>best_threshold,1,0))
confusionMatrix(testdata$predY,testdata$Response,positive="1")
```

```
Confusion Matrix and Statistics

          Reference
Prediction    0    1
         0 1095   57
         1  724  220

               Accuracy : 0.6274
                 95% CI : (0.6063, 0.6481)
    No Information Rate : 0.8678
    P-Value [Acc > NIR] : 1

                  Kappa : 0.1961

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.7942
            Specificity : 0.6020
         Pos Pred Value : 0.2331
         Neg Pred Value : 0.9505
             Prevalence : 0.1322
         Detection Rate : 0.1050
   Detection Prevalence : 0.4504
      Balanced Accuracy : 0.6981

       'Positive' Class : 1
```

K-fold cross validation using caret package

```
library(caret)
kfolds<-trainControl(method="cv",number=4)
kmodel<-train(Response~N_Products+N_Service+BillAmt_1+BillAmt_2,data=traindata,method
="glm",
        family=binomial,trControl=kfolds)

kmodel
```

```
Generalized Linear Model

4894 samples
   4 predictor
   2 classes: '0', '1'

No pre-processing
Resampling: Cross-Validated (4 fold)
Summary of sample sizes: 3671, 3671, 3670, 3670
Resampling results:

  Accuracy   Kappa
  0.8624844  0.00769249
```

```
traindata$pred <- kmodel$finalModel$fitted.values
traindata$predY <-as.factor(ifelse(traindata$pred > best_threshold , 1,0))
confusionMatrix(traindata$predY,traindata$Response,positive="1")
```

```
Confusion Matrix and Statistics

          Reference
Prediction    0    1
         0 2680  138
         1 1566  510

               Accuracy : 0.6518
                 95% CI : (0.6383, 0.6652)
    No Information Rate : 0.8676
    P-Value [Acc > NIR] : 1

                  Kappa : 0.2163

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.7870
            Specificity : 0.6312
         Pos Pred Value : 0.2457
         Neg Pred Value : 0.9510
             Prevalence : 0.1324
         Detection Rate : 0.1042
   Detection Prevalence : 0.4242
      Balanced Accuracy : 0.7091

       'Positive' Class : 1
```

Comment: The Area Under ROC Curve (AUC) for train and test data indicate that the model is stable and performed well on test data. The sensitivity value using optimum threshold is 79% for test data. There is a scope to improve specificity and overall accuracy if more variables are available such as location of the customer, months since first transaction etc.

In the absence of additional variables, the model can be implemented for decision making in next planned campaign.